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Record-breaking ocean temperatures are reported for June, highlighting the continuing effects of climate change and El Niño.

El Niño, a natural climate phenomenon characterized by the periodic warming of the eastern equatorial Pacific Ocean, disrupts the circulation of the atmosphere, bringing great climatic disaster in forms of shifting rainfall, droughts, floods, storm activity, and rising temperatures at the global scale. Now returning in 2026, it has been named "Super El Niño" for its impacts on temperature and environmental conditions as well as global supply.

Introduction

El Niño is a climatic event occurring every 2 to 7 years when ocean waters by the equator off of Peru in the eastern Pacific have greater temperatures than usual, affecting patterns of weather throughout the globe. This is a warm phase of the ENSO (El Niño-Southern Oscillation), where normal upwelling of cold water off South American waters stops with the incoming warmth. Its strength is measured and tracked using sea-surface temperature (SST) anomalies in the Niño 3.4 region (120–170°W, 5°N–5°S), according to the International Research Institute for Climate and Society (IRI).

Likely to become a "Super El Niño", this year's El Niño has reporters and scientists shook by its upcoming impact on the world very soon. According to the NOAA's Climate Prediction Center's (CPC) El Niño advisory, there is an 81% chance that very strong El Niño conditions will appear from October to December that *"would rank among the largest El Niño events in the historical record going back to 1950."* (Mercury News). Furthermore, the same source indicates how there exists a 97% chance of the event persisting through spring 2027, highlighting both the strength and length of this event. The Weather Channel estimates that El Niño could become one of the greatest drivers for increasing rainfall, temperatures, and a 2026 hurricane season of greater volatility, with stronger western Pacific typhoons and weaker eastern Atlantic hurricanes. With 27 other El Niños reported since 1950, and 5 recorded Super El Niños since the first recorded one in 1877.

Overview

Graphs released by The Weather Channel have shown a clear, distant comparison in daily Niño 3.4 sea surface temperatures where June 2026 exceeds the 1991-2020 mean by about 2.5ºF and being above temperatures across all other years after 1982. NOAA estimates that there is a 63% chance that in late summer / early fall that this El Niño can break the 2ºC above average threshold necessary to be regarded as a Super El Niño. This is derived from their temperature anomaly classifications: Weak at 0.5-1.0ºC, Moderate at 1.0-1.5ºC, Strong at 1.5-2.0ºC, and Very Strong at >2.0ºC.

European Centre for Medium-Range Weather (ECMRW) computational models have found that forecasts approaching June have increasingly reporting anomalies through greater deviations present in possible Niño 3.4 temperature rises from the average: from at least a little above 1.0ºC in the April forecast to at least 2.5ºC in the June forecast. Other models have encountered that by the pacific ocean in the equator (off west from the coast of Peru), the sea surface models and estimates a rise of over 2.0ºC implying how a Super El Niño is already approaching based on what such advanced models have deciphered.

Stronger El Niños bring greater effects to the global weather, which can lead to more abnormal winters in terms of climate and location. The southern United States is estimated to be much wetter, with higher chances of snowfall if the air is cold enough. However, the north is warmer and drier during a stronger El Niño. At the global scale, different parts of the world become wetter or drier because of the presence of this El Niño. Parts of Africa, the Caribbean, the Philippines, Indonesia, Australia, and the north of South America are areas that have greater dryness. On the other hand, the southern United States, parts of eastern Africa, and some South American countries in the south have typically greater wetness in this climatic event.

As for current conditions in early-mid July, the El Niño 3.4 index sat at +1.2ºC and rising. With multiple models reporting this is expected to further intensify, there are specific individuals who have already made their own estimations on what will happen during the peak of El Niño in autumn-winter. Zeke Hausfather at the Global Commons Alliance forecasts around a 3.6ºC peak above the national average, while Daniel Swain from Mercury News sees a 3.0-3.5ºC peak. Even with the strongest recent events peaking at 2.5ºC and 2.75ºC, these forecasts exceed whatever past record there was by such a wide margin.

Perspectives

Weather forecasting firms such as the Weather Channel, NOAA/CPC, and the IRI have all come together in emphasizing such mechanisms and confidence levels with the strength of this El Niño. Westerly wind bursts pushing the warmth of the water towards the east, along with a

downwelling Kelvin wave that is deepening the thermocline. With no models having been exposed to this strength before, forecasters have already shown their perspective through labeling this as “uncharted territory”.

In an economic standpoint, El Niño is expected to bring significant increases to global food commodity prices lasting into 2028. An extreme El Niño such as this adds to the already inflated costs from household living and the pressure with an inflation shock. Specifically, Goldman Sachs reported a 15.8% jump in said prices along with eurozone food up to 1.3% which highlights the impact of the double shock that is the Iran war three-year high food prices along with the effects of climate change through the abnormal El Niño adding to this harm food prices to extreme levels. Risilience, a climate analytics firm, encountered a 14.3% hit to total global agricultural output, equivalent to $342 billion in production losses with price shocks of 50-100% for the most exposed of food commodities including rice, sugar, palm oil, and coffee.

Popular Mechanics released an article comparing this El Niño with the devastating great famine of 1876-1878, with the same climatic event contributing to the deaths of 50 million across Asia, Africa, and South America, (with more than 6 million in India alone). However, there are important caveats to be considered when comparing similar events to each other such as articles such as these comparing through death toll. There were political and economic factors that drove the deaths of these 50 million even further other than just the environmental crisis. With ENSO tracking systems being already developed, there is no trace of a large-scale famine coming soon despite the statistically stronger El Niños we’ve had over the past few years and decades.

Implications

Despite how severe this El Niño has been estimated to become based on reports from the most widely known and advanced climate forecasting, statistical, and analytic firms, this is a regional situation. What this implies is that certain locations of the world will become drier or wetter. It doesn’t worsen the climate, but alters it to be abnormal such as reshaping rainfall in certain locations rather than developing stronger rainfall all throughout the globe, for example. A drier monsoon in India in 2026 already showed evidence surrounding the effects El Niño is having on certain regions further than others.

Studies have shown that El Niño actually produces strong climate feedback. In this case, because global temperatures lag El Niño by several months, this could lead to much of the added heat becoming prevalent in 2027. Hence, sources suggest this could be the hottest year on record which brings more concern to what future environmental events such as these can add on to an already negatively changing climate worldwide. Humanity must be responsible for what is coming soon as the 50 million losses in the El Niño of 1876-1878 was due to issues in politics and economies, which no longer exist. With warning systems, statistical analysis, and

atmospheric understanding, populations must be supported to prevent a humanitarian disaster from occurring.

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